

Chengshuai (Abraham) Yang, Ph.D.

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[Google Scholar](#) · [research.platformai.org](#) · [ai-level.platformai.org](#) · [github.com/integritynoble](#)

Summary

I build AI agents that do scientific work, and the instruments that decide whether the work was actually done. Ten years in computational imaging and inverse problems (Physical Review Letters, Optica, ECCV, IJCV; 762 citations, h-index 17) now applied to a harder question: when an autonomous agent reports a finished result, how do you know it is real? I designed and ran **AI-Level**, a public measurement standard whose score *falls* when a model reports work it did not do — false completion is priced, not ignored — and built the agent harnesses it measures. I am looking for a role designing AI agents that carry out scientific research.

What I Work On

- **Agents that do research.** Harness design: planning, motor execution, external verification, and durable memory across restarts, applied to imaging, medical physics and drug design.
- **Measuring delivered work, not benchmark scores.** Delegation under bounded human intervention; the false-completion / false-rejection pair; verified delivery as the primitive.
- **Making AI-for-science results machine-checkable.** Verifiers an agent cannot write to, preregistered thresholds, and evidence an outside party can re-derive.
- **Physics-grounded inverse problems.** Forward-operator structure, model mismatch, calibration, and reconstruction under uncertainty.

Experience

Research Director & Co-Founder — NextGen PlatformAI C Corp

Mar 2024 – Present

San Jose / Los Angeles, CA

- Designed **AI-Level**, a measurement standard for how much verified work a model can complete as a standardized agent. Public board, five models measured end to end, every row re-derivable from its stored record.
- Built **AI-Level Bench**: canonical test laws with machine-checked admissibility, a reference harness family (HG0–HG2), public/private splits with a committed salt, and hidden verifiers outside the agent’s write set.
- Built a multi-agent research fleet (imaging, medical physics, drug design, cancer, virtual cell) on one harness, with per-agent memory, role separation and external acceptance.
- Demonstrated restart-recovery measurement on a live agent: state written by one process recovered by another after termination, against a benchmark-authored control that fails it.
- Ran the whole programme on self-funded compute; ten of the works below were produced in this period.

Research Associate — UT Southwestern Medical Center

Apr 2026 – Present

Dallas, TX

- Derived a training-time hemoglobin optical prior from light transport and showed it improves vascular and mucosal lesion classification in *in vivo* human wireless capsule endoscopy.
- **Selected for oral presentation, World Molecular Imaging Congress 2026** — five blinded peer reviewers, scored on scientific merit.

Machine Learning Engineer — Workmagic

Oct 2023 – Mar 2024

San Francisco, CA

- Diffusion-based generative video for advertising; data curation, evaluation and error-analysis pipelines. Prototyped an internal LLM product copilot.

Postdoctoral Scholar — University of California, Los Angeles (ECE)

Oct 2021 – Sep 2023

Los Angeles, CA

- Diffusion and GAN priors for light-field tomography and hyperspectral imaging; reconstruction as Bayesian inference under learned priors; reproducible multi-GPU experiment pipelines.

Machine Learning Optics Team Leader — Westlake University

Oct 2020 – Sep 2021

Hangzhou, China

- Founded and led a computational imaging + ML group (5 Ph.D. students and RAs); ECCV 2022 and IJCV 2023 on scalable deep unfolding and adaptive plug-and-play reconstruction.

Education

Postdoctoral Training, UCLA — Machine Learning; Optical/Biomedical Imaging	2021 – 2023
Ph.D., East China Normal University — Optical Imaging; Machine Learning	2015 – 2020
B.S., Nantong University — Applied Physics; Computer Science	2011 – 2015

Research Programme, 2024–2026

Twenty-six works, all at research.platformai.org. Status is stated on every entry: one preprint, one submitted manuscript, the remainder working drafts. None has been peer reviewed, and the archive says so before the first paper.

Measuring agent-realized intelligence

- *Unified Intelligence Theory and the AI-Level* (v2.4, 64 pp., with T. Xue) — coordinates $[C, I, O, T, H, SA]$ with memory reported beside I ; a gated hierarchy that refuses to average.
- *AI-Level Bench* (v0.1, 55 pp.) — one instrument, two subjects: an agent as the model–harness pair it is, and a bare model through reference harnesses.
- *The AI-Level Index v2.0* (23 pp., with T. Xue) — a difficulty-calibrated, non-saturating scale; bands rather than ranks; preregistered reliability rules.

Agents that do science

- *One Method for AI4Science* (40 pp., with T. Xue) — from computational imaging to virtual cells and drug design on one method.
- *The AI4Science Layer* (v0.1) and *The Domain Agent Library* — six domain agents instantiated on a single general harness by manifest.

Research integrity for AI-produced results

- *Making AI-for-Science Results Machine-Checkable* (with T. Xue) — the missing floor and a four-layer scheme.
- *Physics-Grounded Silent Failure Detection for AI Research Agents*.

Computational imaging foundations

- *Eleven Primitives and Three Gates* (97 pp., with D. J. Brady and X. Yuan) — preprint.
- *InverseNet* (33 pp.) — operator mismatch and calibration benchmark; under review.
- *The Finite Primitive Basis Theorem* — arXiv:2602.20550.

Selected Peer-Reviewed Publications

762 citations, h-index 17, i10-index 19.

- Yang, C.; Zhang, S.; Yuan, X. Ensemble Learning Priors Driven Deep Unfolding for Scalable Video Snapshot Compressive Imaging. **ECCV** (2022).
- Wu, Z.; **Yang, C.**; Su, X.; Yuan, X. Adaptive Deep Plug-and-Play Algorithm for Video Snapshot Compressive Imaging. **IJCV** 131(7): 1662–1679 (2023).
- Yang, C.; Cao, F.; Qi, D.; et al. Hyperspectrally Compressed Ultrafast Photography. **Physical Review Letters** 124, 023902 (2020).
- Yang, C.; Qi, D.; et al. Optimizing Codes for Compressed Ultrafast Photography by the Genetic Algorithm. **Optica** 5(2): 147–153 (2018).
- Yang, C.; et al. High-Fidelity Image Reconstruction for Compressed Ultrafast Photography. **Photonics Research** 9(2): B30–B40 (2021).
- Zhao, R.; **Yang, C.**; Smith, R.T.; Gao, L. Coded Aperture Snapshot Spectral Imaging Fundus Camera. **Scientific Reports** 13 (2023).
- Du, X.; et al. Label-Free Hyperspectral Imaging and Deep-Learning Prediction of Neuronal Functional Dynamics. **PNAS Nexus** (2022).

Selected Presentation

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- **Oral presentation**, World Molecular Imaging Congress **2026**. *A training-time hemoglobin optical prior derived from light transport improves vascular and mucosal lesion classification in in vivo human wireless capsule endoscopy*. Abstract 142; selected from blinded review by five peer reviewers.

Open Instruments

- ai-level.platformai.org — the AI-Level index and public board; every row reproducible from its record with one command.
- research.platformai.org — the research archive, with status stated on every entry.
- github.com/integritynoble — Physics World Model, AI-Level Bench, agent harnesses.

Honors

- Outstanding Ph.D. Award, East China Normal University (2020).
- Future Scientist Development Program Fellow, ECNU (2019); Academic Innovation Promotion Award (2018).

Technical

Languages & tools: Python, C/C++, MATLAB, CUDA, SQL; Linux, Git, Docker; PyTorch, TensorFlow.

Agents & LLMs: multi-agent harness design, tool/permission contracts, evaluation and benchmark construction, memory and restart-recovery, LLM provider integration at scale.

Methods: inverse problems, optimization, plug-and-play and unrolled methods, diffusion priors, uncertainty quantification, reproducible multi-GPU experimentation.

Human languages: English (professional), Chinese (native).